

Ritwij Aryan Parmar

Application for Founding Software Engineer, Robotics Infrastructure
New York, NY | Open to San Francisco onsite | Available full-time

Role	GitHub	LinkedIn	Resume	Transcript
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Why Lambda Robotics

Chen asked me on LinkedIn to apply through YC. What drew me to this opening is that it is not limited to autonomy: it is the infrastructure that lets real robots be registered, deployed, monitored, updated, and recovered across different customer sites. That is the direction my work has naturally been moving toward.

Relevant systems experience

- **Tata Elxsi:** Migrated an autonomous-driving stack from ROS1 to ROS2 and tuned DDS QoS in CARLA, reducing inter-module latency by 40% to under 100 ms.
- Built a pipeline that synchronized RGB, depth, semantic labels, ego pose, and calibration data, then mined failures across 120+ difficult simulation runs.
- **UB DRONES Lab:** Built a ROS2 visual SLAM pipeline for Boston Dynamics Spot and a real-robot inspection workflow for field data.
- Built backend and deployment systems with Spring Boot, RabbitMQ, Docker, FastAPI, GitHub Actions, and GCP, including a GPU inference service with runtime metrics and performance diagnostics.

Merged open-source infrastructure work

- [TraceRoot #1134](#) - Kept live trace streams from silently losing late spans and added 98 focused tests.
- [GreptimeDB #8341](#) - Streamed Prometheus label discovery instead of loading every catalog table first.
- [Unslloth #7191](#) - Added a token-protected control plane with validated APIs across 11 tools.
- [Daft #7061](#) - Corrected OpenTelemetry resource configuration without overriding upstream settings.
- [Firecrawl #3671](#) - Fixed a Redis client lifecycle leak with shared-client reuse and clean shutdown handling.

Across this work, I have repeatedly dealt with failures at system boundaries: devices and services, telemetry and storage, deployment configuration and runtime behavior, and control APIs and authentication. I would be excited to bring that ownership to Lambda and help build the fleet platform needed to operate robots reliably across real sites.

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EXPERIENCE

- Distributed Robotics and Networked Embedded Sensing (DRONES) Lab** Buffalo, NY
Research Aide – Robotics Systems Jul 2025 – Feb 2026
 - Reduced mapping drift to under **1.2%** across **500 meters** of GPS-denied environments by building a **ROS2 visual SLAM** pipeline for Boston Dynamics Spot with Gaussian Splatting integration.
 - Improved **LiDAR and VIO fusion** for low-texture subterranean navigation, reducing trajectory estimation error by **30%** while sustaining **20 hertz** real-time state estimation.
 - Built a real-robot inspection workflow for culvert-assessment data, converting field videos into defect masks and reviewable polygon annotations across **1,400+** sampled frames with per-run IoU/F1 reports.
- Tata Elxsi – Global automotive engineering firm** Bengaluru, India
Software Engineer Intern Jan 2024 – Jun 2024
 - Migrated an autonomous vehicle software stack from **ROS1 to ROS2** and tuned **DDS QoS** behavior in CARLA, reducing inter-module latency by **40%** and achieving sub-100 millisecond communication latency.
 - Built an **Extended Kalman Filter** vehicle state estimator that improved positioning accuracy by **35%**, and implemented a safety-constrained route planner with deterministic state transitions and bounded-latency path generation.
 - Built an autonomy evaluation-data pipeline that converted simulator runs into detector-ready datasets, aligning RGB frames, depth maps, semantic labels, ego pose, and calibration metadata into validated 2D/3D annotation packages.
 - Automated failure mining across **120+** night, rain, occlusion, cut-in, and pedestrian-crossing simulation runs by flagging low-confidence detections, missed ground-truth matches, and label/calibration mismatches.
- LLMate.ai – Seed-stage GenAI analytics startup** Bengaluru, India
Software Engineer Intern Apr 2023 – Jul 2023
 - Built backend services with Spring Boot, RabbitMQ, Docker, and GitHub Actions for production data workflows, reducing p95 response latency by about **40%**.
 - Built a natural-language SQL backend over **50,000+** records with parser validation, read-only execution, row/time limits, query-cost controls, and result caching.

PROJECTS

- HelixServe – Low-Latency GPU Inference Server – GitHub** – Increased serving throughput **5.7x** from **175.8 to 1,007.8 tokens/sec** and reduced p95 latency **75%** from **3.28s to 0.80s** on GCP NVIDIA L4 by implementing paged KV cache, continuous batching, prefix caching, CUDA Graph decode, and runtime metrics.

OPEN-SOURCE ENGINEERING

- Robotics/Perception and Data Systems – GitHub** – Contributed **30+ merged upstream PRs** across computer-vision evaluation, robotics-style telemetry/data tooling, vector search, observability databases, and AI infrastructure.
- Roboflow Supervision, FiftyOne, Daft, Qdrant, GreptimeDB** – Fixed non-square OBB IoU in Roboflow Supervision and added configurable keypoint-evaluation behavior to FiftyOne; also improved Daft MCAP URL handling, Qdrant scalar-quantized L2 scoring, and GreptimeDB Prometheus metadata discovery.

SKILLS

Languages: Python, C++, Java, SQL

Robotics: ROS2, ROS1, SLAM, LiDAR/VIO Sensor Fusion, EKF, CARLA, DDS QoS, State Estimation, Path Planning, Robot Bring-up

Perception/Data: OpenCV, Camera/LiDAR Data Processing, 2D/3D Annotation Validation, Perception Metrics, Failure Mining, Sensor Calibration Metadata

Software: Linux, Git, Docker, FastAPI, Spring Boot, RabbitMQ, GCP, Cloud Run, CI/CD

EDUCATION

- University at Buffalo – SUNY** Buffalo, NY
Master of Science in Computer Science and Engineering; GPA: 3.74/4.0 Jan 2025 – May 2026
- Manipal Institute of Technology** Manipal, India
Bachelor of Technology in Computer Science and Engineering Aug 2020 – Aug 2024

Program

Graduate Career
Engnr & Appl Sci Masters Program
Computer Sci And Engr MS Major

Requirement Term

Spring 2025
Spring 2026
Spring 2026

COURSE HISTORY & COURSES NOT ALLOCATED

COURSE HISTORY - (This includes a list of all coursework taken in your Graduate Career) (RG-0011).

Course History

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Spr 2025	CSE 531LEC	Algorithms Anal & Dsgn 1	B	3.00	EN
Spr 2025	CSE 560LEC	Data Models Query Lang	A	3.00	EN
Spr 2025	CSE 574LEC	Intro Machine Learning	A	3.00	EN
Spr 2025	CSE 587LEC	Data Intensive Computing	A	3.00	EN
Sumr 2025	CSE 676LEC	Deep Learning	A	3.00	EN
Fall 2025	CSE 521LEC	Operating Systems	B+	3.00	EN
Fall 2025	CSE 565LEC	Computer Security	A-	3.00	EN
Fall 2025	CSE 700TUT	Independent Study	S	3.00	EN
Fall 2025	EAS 507LEC	Statistical Data Mining II	A	3.00	EN
Spr 2026	CSE 635LEC	NLP and Text Mining	A-	3.00	EN

COURSES NOT ALLOCATED

COURSES NOT ALLOCATED - This list includes all courses that are not being applied towards your graduate program requirements.(RQ-100225)

OK **COMPUTER SCIENCE M.S.**
Satisfied: COMPUTER SCIENCE M.S. - 30 Units Required (2026-Present) (RG-012075)

OK **REQUIRED CORE COURSES**
Satisfied: REQUIRED CORE COURSES - 12 Units Required (RQ-102367)
(Flex program) Minimum B (3.0) grade average required across the core courses.

Analysis of Algorithms

Satisfied: Analysis of Algorithms - 1 Course Required (LN-010)

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Spr 2025	CSE 531LEC	Algorithms Anal & Dsgn 1	B	3.00	EN

Introduction to Machine Learning

Satisfied: Introduction to Machine Learning - 1 Course Required (LN-020)

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Spr 2025	CSE 574LEC	Intro Machine Learning	A	3.00	EN

Introduction to Operating Systems

Satisfied: Introduction to Operating Systems - 1 Courses Required (LN-030)

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Fall 2025	CSE 521LEC	Operating Systems	B+	3.00	EN

Computer Security

Satisfied: Computer Security - 1 Course Required (LN-040)

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Fall 2025	CSE 565LEC	Computer Security	A-	3.00	EN

VERIFY: Core Courses GPA Requirement

Satisfied: Core Courses GPA Requirement - 3.0 GPA Required (LN-050)

GPA: 3.000 required, 3.500 completed

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Spr 2025	CSE 531LEC	Algorithms Anal & Dsgn 1	B	3.00	EN
Spr 2025	CSE 574LEC	Intro Machine Learning	A	3.00	EN
Fall 2025	CSE 521LEC	Operating Systems	B+	3.00	EN
Fall 2025	CSE 565LEC	Computer Security	A-	3.00	EN

Choose elective courses based on your interests or to bolster your specialization.
Maximum of 3 units CSE 598 Internship;
Maximum 6 credits outside of CSE

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Fall 2025	EAS 507LEC	Statistical Data Mining II	A	3.00	EN
Sumr 2025	CSE 676LEC	Deep Learning	A	3.00	EN
Fall 2025	CSE 700TUT	Independent Study	S	3.00	EN

OK **CS MS SPECIALIZATION COURSES**
Satisfied: COMPUTER SCIENCE MS SPECIALIZATION COURSES - 6 Units Required (RQ - 102368)
(Flex program)
Select from a range of specializations as detailed on the CSE website, or build your own.
Take two courses here toward a specialization.

CS MS SPECIALIZATION COURSES

Satisfied: COMPUTER SCIENCE MS SPECIALIZATION COURSES - 6 Units Required (RQ - 102368)
Select from a range of specializations as detailed on the CSE website, or build your own.
Take two courses here toward a specialization.

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Spr 2025	CSE 560LEC	Data Models Query Lang	A	3.00	EN
Spr 2025	CSE 587LEC	Data Intensive Computing	A	3.00	EN

CSE 676LEC has been directed to this line.
CSE 676 and CSE 700 will be used as an elective and not a capstone as per the DGS.

CSE 700TUT has been directed to this line.
CSE 676 and CSE 700 will be used as an elective and not a capstone as per the DGS.

OK **VERIFY CAPSTONE PROJECT COMPLETION**
Satisfied: CULMINATING EXPERIENCE - PROJECT - (RQ-102370)
(Flex program)

Will satisfy with successful completion of an embedded Project Capstone course or with completion of a project in CSE 700* and completion of the MS project milestone.

CSE 700 requires advisor approval.

MS Project Embedded Courses (Verify)

Project Embedded Courses: CSE 562, 573, etc - 3 Units Required (LN-010)

*Culminating Experience Course, a minimum grade of B- or higher is required

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Sumr 2025	CSE 676LEC	Deep Learning	A	3.00	EN
Spr 2026	CSE 635LEC	NLP and Text Mining	A-	3.00	EN

OK **ELECTIVES & PROJECT CAPSTONE REGISTRATION**
Satisfied: ELECTIVES & PROJECT CAPSTONE REGISTRATION - 12 Units Required (RQ-102369)
(Flex program)

CS MS Project Capstone Course Options

Satisfied: CS MS Project Capstone Course Options - 3 Units Required (LN-010)
Register for a project capstone course in your final semester and complete a master's project aligning with your specialization.
Minimum grade of B- or higher is required

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Spr 2026	CSE 635LEC	NLP and Text Mining	A-	3.00	EN

Electives

Satisfied: Electives - 9 Units Required (LN-030)

OK **DEGREE GPA & UNIT VERIFICATION**
Satisfied: DEGREE GPA & UNIT VERIFICATION - 3.0 GPA / 30 Units Required (RQ-900000301)

Degree GPA & Unit Verification

Satisfied: Degree GPA & Unit Verification - 3.0 GPA / 30 Units Required (LN-010)

Units: 30.00 required, 30.00 used
GPA: 3.000 required, 3.741 completed

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Spr 2025	CSE 531LEC	Algorithms Anal & Dsgn 1	B	3.00	EN
Spr 2025	CSE 560LEC	Data Models Query Lang	A	3.00	EN
Spr 2025	CSE 574LEC	Intro Machine Learning	A	3.00	EN
Spr 2025	CSE 587LEC	Data Intensive Computing	A	3.00	EN
Sumr 2025	CSE 676LEC	Deep Learning	A	3.00	EN
Fall 2025	CSE 521LEC	Operating Systems	B+	3.00	EN
Fall 2025	CSE 565LEC	Computer Security	A-	3.00	EN
Fall 2025	CSE 700TUT	Independent Study	S	3.00	EN
Fall 2025	EAS 507LEC	Statistical Data Mining II	A	3.00	EN
Spr 2026	CSE 635LEC	NLP and Text Mining	A-	3.00	EN

CSE ALL COURSES GPA

All CSE courses taken at the graduate level are included in this GPA. (RQ-000102088)

ALL CSE Courses GPA

ALL CSE Courses GPA - (LN-010)

Units: 27.00 used

GPA: 3.708 completed

Courses Used

Term	Course #	Course Title	Grade	Units	Type
Spr 2025	CSE 531LEC	Algorithms Anal & Dsgn 1	B	3.00	EN
Spr 2025	CSE 560LEC	Data Models Query Lang	A	3.00	EN
Spr 2025	CSE 574LEC	Intro Machine Learning	A	3.00	EN
Spr 2025	CSE 587LEC	Data Intensive Computing	A	3.00	EN
Sumr 2025	CSE 676LEC	Deep Learning	A	3.00	EN
Fall 2025	CSE 521LEC	Operating Systems	B+	3.00	EN
Fall 2025	CSE 565LEC	Computer Security	A-	3.00	EN
Fall 2025	CSE 700TUT	Independent Study	S	3.00	EN
Spr 2026	CSE 635LEC	NLP and Text Mining	A-	3.00	EN